

Note 1.2 Application of Propositional Logic

Introduction

- This section introduces some important applications to mathematics, computer science and other disciplines.

Translating English Sentences

- **Applications:**
 - Remove the ambiguity in natural languages
 - Convenient to analyze and infer.
- **Strategy:**

Transform the sentence into proposition logic.

System Specifications

- **Application:**
 - Produce precise specification to be used as the basis for system development.
 - Ensure the system specification be consistent.(A list of propositions is consistent if it is possible to assign truth values to the proposition variables so that each proposition is true)

Boolean Searches

- **Definition:**

The searches employing the techniques from propositional logic.

Logic Puzzles

- **Definition:**
 - The puzzles can be solved using logical reasoning.

Logic Circuits

- **Definition:**

A logic circuit (or digital circuit) receives input signal each a bit and produces output signals each a bit(0 or 1). Gates is the basic circuits in the circuit.
- **Category of Gate:**
 - Inverter(not gate): take the input p and produce as output $\neg p$.
 - OR gate: take the inputs p and q and produce as output $p \vee q$.

- AND gate: take the inputs p and q and produce as output $p \wedge q$.



Inverter



OR gate



AND gate